

Investigating a function with the derivative

Four lessons on the full procedure — sign of the derivative, stationary points, their nature, and the sketch that results.

LESSON 19–22

4 × 40 MIN

ALGEBRA 11, §1.7

P1 · 7.7–7.8

LESSON CLOCK

16

24'

30'

30'

40'

40'

The sign of the derivative	16 min
Stationary points	24 min
Classifying them	30 min
The second derivative test	30 min
Full sketches	40 min
Practice and homework	40 min

LEARNING OBJECTIVES

- Use the sign of f' to find intervals of increase and decrease.
- Find stationary points by solving $f'(x) = 0$.
- Classify each stationary point by the sign change or by f'' .
- Sketch a curve using intercepts, stationary points and behaviour at the ends.

TEXTBOOK

National	pp. 91–110
Cambridge	Pure Mathematics 1, pp. 168–185
Workbook	P1 Exercise 7G, 7H

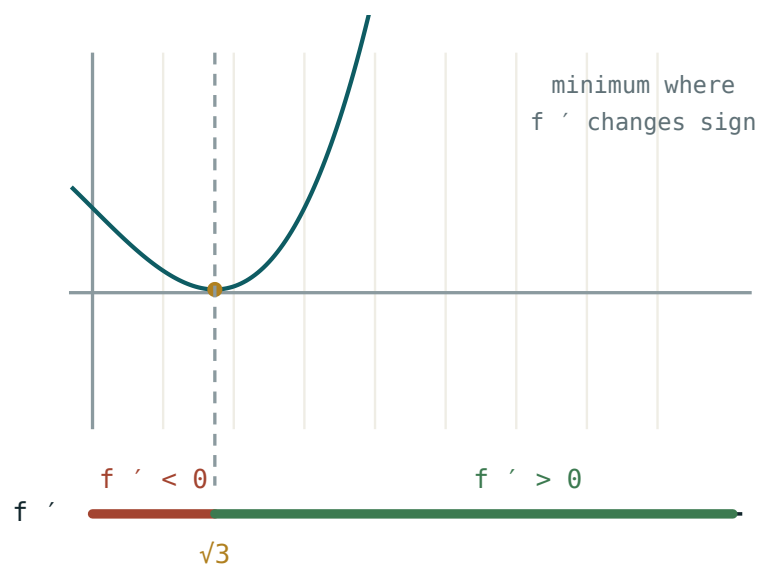
01

Explanation

The sign of the derivative

THE LINK THAT MAKES CALCULUS USEFUL

$f'(x) > 0$ on an interval $\implies f$ is **increasing** there. $f'(x) < 0 \implies f$ is **decreasing**. $f'(x) = 0 \implies$ the tangent is horizontal — a **stationary point**.



Where the derivative is above the axis, the curve climbs. Where it is below, the curve falls.

So the shape of f is read entirely off the **sign** of f' , and the sign of f' is read off its factorised form.

Finding and tabulating

Example. $f(x) = x^3 - 3x^2 - 9x + 5$.

$$f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$$

x	< -1	-1	$-1 \text{ TO } 3$	3	> 3
f'	+	0	−	0	+
f	↗	max	↘	min	↗

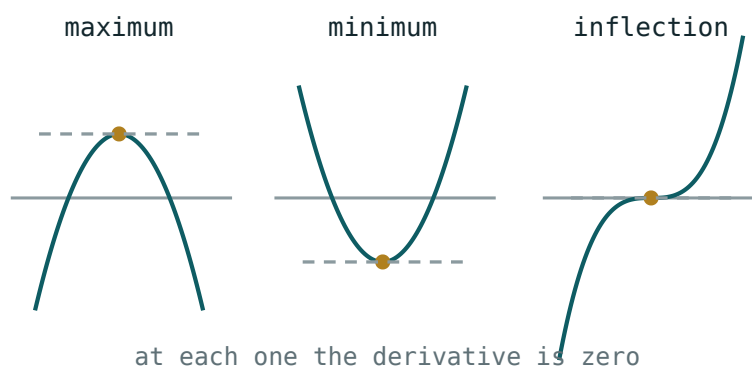
Local maximum at $(-1, 10)$, local minimum at $(3, -22)$.

ALWAYS FACTORISE f'

An unfactorised quadratic hides its sign. The factorised form gives the roots and the sign chart in one step.

Classifying: two tests

TEST	MAXIMUM	MINIMUM	INCONCLUSIVE WHEN
sign of f'	+ then -	- then +	never
$f''(x_0)$	$f'' < 0$	$f'' > 0$	$f'' = 0$



A maximum, a minimum and a stationary point of inflection. All three have a horizontal tangent.

$f''(x_0) = 0$ PROVES NOTHING

$y = x^4$ has $f''(0) = 0$ and a minimum; $y = x^3$ has $f''(0) = 0$ and an inflection. When the second-derivative test is silent, go back to the sign of f' .

A **stationary point of inflection** has $f' = 0$ but **no** sign change: the curve flattens and carries on in the same direction, as $y = x^3$ does at the origin.

The full sketch

SIX STEPS, IN ORDER

1 domain · **2** intercepts ($x = 0$, and $f(x) = 0$ if it factorises) · **3** f' , factorised · **4** stationary points and their nature · **5** behaviour as $x \rightarrow \pm\infty$ · **6** draw, plotting the stationary points first.

For a cubic with a positive leading coefficient, step 5 says the curve comes up from $-\infty$ on the left and goes to $+\infty$ on the right. Together with the two stationary points, the shape is completely determined.

Second derivative and concavity. $f'' > 0$ means the curve bends upwards (concave up, holding water); $f'' < 0$ means it bends downwards. Where f'' changes sign there is a **point of inflection** — which need not be stationary.

02 Worked examples

EXAMPLE 1 Investigate $f(x) = x^3 - 12x + 5$ fully.

① $f' = 3x^2 - 12 = 3(x - 2)(x + 2)$

② Stationary at $x = \pm 2$.
 $f(-2) = 21, f(2) = -11$

③ $f'' = 6x$
 $f''(-2) = -12 < 0$ max; $f''(2) = 12 > 0$ min

④ Increasing on $x < -2$ and $x > 2$.
 Decreasing between.

ANSWER

max $(-2, 21)$, min $(2, -11)$

EXAMPLE 2 Show $f(x) = x^3 + 3x$ has no stationary points.

① $f' = 3x^2 + 3$

② $3x^2 + 3 = 0 \Rightarrow x^2 = -1$
 No real solution.

③ $f' > 0$ everywhere.

ANSWER

None — the function increases on all of $\mathbb{R}$

EXAMPLE 3 Classify the stationary point of $f(x) = x^4 - 4x^3$ at $x = 0$.

① $f' = 4x^3 - 12x^2 = 4x^2(x - 3)$

Stationary at 0 and 3.

② $f'' = 12x^2 - 24x, f''(0) = 0$

Test inconclusive.

③ Sign of f' near 0: $4x^2 \geq 0, (x - 3) < 0$.

Negative on both sides.

④ No sign change.

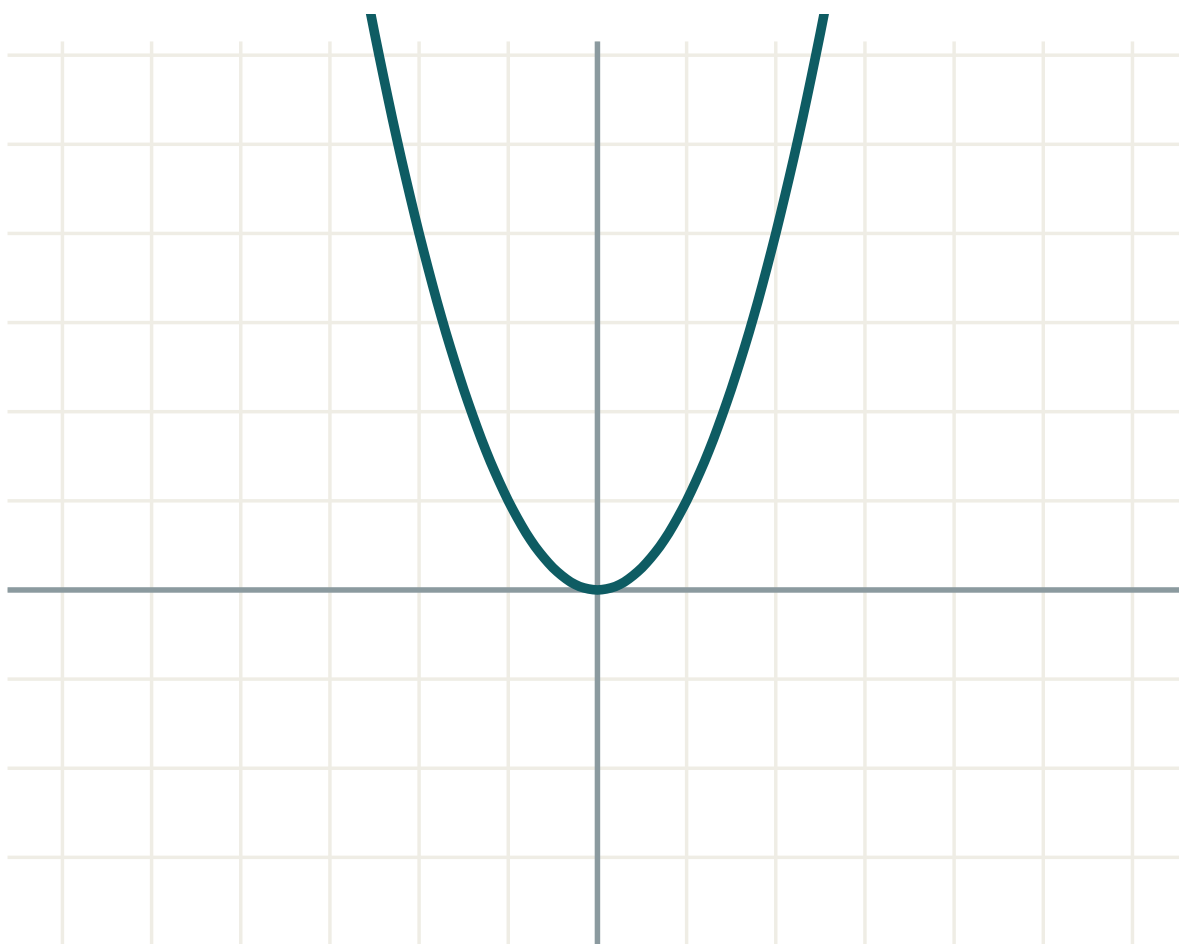
ANSWER

A stationary point of inflection at $(0, 0)$

03 Interactive model

Show the sign chart and the sketch side by side — every row of the table is a piece of the curve.

Shape from the derivative



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Increasing function	O'suvchi funksiya	Возрастающая функция
Decreasing function	Kamayuvchi funksiya	Убывающая функция
Stationary point	Statsionar nuqta	Стационарная точка
Critical point	Kritik nuqta	Критическая точка
Local maximum	Lokal maksimum	Локальный максимум
Local minimum	Lokal minimum	Локальный минимум
Point of inflection	Egilish nuqtasi	Точка перегиба
Second derivative	Ikkinchi hosila	Вторая производная
Sign chart	Ishoralar jadvali	Таблица знаков
Curve sketching	Grafik chizish	Построение графика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$f'(x) > 0$ means the function is:

positive

increasing

concave up

stationary

A stationary point has:

$f = 0$

$f' = 0$

$f'' = 0$

$f' > 0$

$f' = 0$ and $f'' < 0$ gives a:

minimum

maximum

inflection

root

If $f''(x_0) = 0$ the point is:

 a maximum

 a minimum

 an inflection

 not decided by this test

Stationary points of $y = x^3 - 12x$ are at:

 $x = 0$
 $x = \pm 2$
 $x = \pm 4$
 $x = 12$

$y = x^3$ at $x = 0$ has:

 a maximum

 a minimum

 a stationary inflection

 no stationary point

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Stationary points of $y = x^2 - 6x$

ANSWER $x = 3$

2. Classify it

ANSWER minimum

3. Stationary points of $y = 4x - x^2$

ANSWER $x = 2$, maximum

4. Is $y = 3x + 1$ increasing?

ANSWER yes, everywhere

5. Stationary points of $y = x^3$

ANSWER $x = 0$

6. Classify it

ANSWER stationary inflection

7. f'' for $y = x^3 - 3x$

ANSWER $6x$

Medium · the standard question

7 PROBLEMS

1. Investigate $y = x^3 - 3x$

ANSWER $\max(-1, 2), \min(1, -2)$

2. Investigate $y = x^3 - 12x + 5$

ANSWER $\max(-2, 21), \min(2, -11)$

3. Investigate $y = 2x^3 - 3x^2 - 12x$

ANSWER $\max(-1, 7), \min(2, -20)$

4. Where is $y = x^4 - 8x^2$ decreasing?

ANSWER $x < -2$ and $0 < x < 2$

5. Stationary points of $y = x^4 - 8x^2$

ANSWER $x = 0, \pm 2$

6. Nature of each

ANSWER 0 max; ± 2 minima

7. Point of inflection of $y = x^3 - 6x^2$

ANSWER $(2, -16)$

Hard · stretch and reason

7 PROBLEMS

1. Investigate $y = \frac{x}{x^2 + 1}$

ANSWER max (1, 0.5), min (-1, -0.5)

2. Investigate $y = x + \frac{1}{x}$

ANSWER max (-1, -2), min (1, 2)

3. Find a so $y = x^3 + ax$ has no stationary point

ANSWER $a > 0$

4. Find a, b so $y = x^3 + ax^2 + b$ has a stationary point at (2, 3)

ANSWER $a = -3, b = 7$

5. Classify the stationary point of $y = x^4$ at 0 and explain why f'' fails

ANSWER Minimum; $f''(0) = 0$

6. Show $y = x^3 + x$ is increasing everywhere

ANSWER $f' = 3x^2 + 1 > 0$

7. Sketch $y = x^3 - 3x^2 + 4$ fully

ANSWER max (0, 4), min (2, 0), root at $x = -1$

07 Homework

Homework — 6 tasks

Every investigation needs the sign chart, not just the answers.

- Investigate $y = x^3 - 6x^2 + 9x$ fully and sketch it.
- Investigate $y = x^4 - 2x^2$ fully and sketch it.
- Find the intervals on which $y = 2x^3 + 3x^2 - 36x$ increases.
- Find and classify the stationary points of $y = x + \frac{4}{x}$.

5. Find a and b so that $y = x^3 + ax^2 + bx$ has stationary points at $x = 1$ and $x = 3$.
6. Explain, with two examples, why $f''(x_0) = 0$ does not classify a stationary point.